

1 Mean-Value Theorem for Multivariable Functions

• Recall that for a real-valued one-variable differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, we have the **mean-value theorem**, which states that

$$f(x_1) - f(x_2) = f'(x_0)(x_1 - x_2),$$

where x_0 lies between x_1 and x_2 . Unfortunately, this equation cannot be generalized to the vector-valued multivariable case. From now on, for two points $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, we denote

$$l(\mathbf{x}; \mathbf{y}) := \{t\mathbf{x} + (1-t)\mathbf{y} \mid t \in [0, 1]\},$$

i.e., $l(\mathbf{x}; \mathbf{y})$ means the line segment from $\mathbf{x}$ to $\mathbf{y}$.

The following theorem gives the correct form of mean-value theorem for multivariable functions.

Theorem 1. (Mean-Value Theorem for Multivariable Functions) Let $S \subseteq \mathbb{R}^n$ be an open subset of $\mathbb{R}^n$ and let $f : S \rightarrow \mathbb{R}^m$ be a differentiable function on S . Given two different points $\mathbf{x}$ and $\mathbf{y}$, suppose $l(\mathbf{x}; \mathbf{y}) \subseteq S$, then for any vector $\mathbf{a}$ in $\mathbb{R}^m$, there exists a point $\mathbf{z} \in l(\mathbf{x}; \mathbf{y})$ such that

$$\mathbf{a} \cdot (f(\mathbf{y}) - f(\mathbf{x})) = \mathbf{a} \cdot (f'(\mathbf{z})(\mathbf{y} - \mathbf{x})). \quad (1)$$

Proof: Denote $\mathbf{u} := \mathbf{y} - \mathbf{x}$. Since S is open and $l(\mathbf{x}; \mathbf{y}) \subseteq S$, $\exists \delta > 0$ such that

$$\{t\mathbf{x} + (1-t)\mathbf{y} \mid t \in (-\delta, 1+\delta)\} \subseteq S.$$

Now, for any given vector $\mathbf{a}$ in $\mathbb{R}^m$, define the real-valued one-variable function $F : (-\delta, 1+\delta) \rightarrow \mathbb{R}$ as

$$F(t) := \mathbf{a} \cdot f(\mathbf{x} + t\mathbf{u}).$$

It's clear that $F(t)$ is differentiable and by the definition of F , we get

$$F'(t) = \mathbf{a} \cdot D_{\mathbf{u}}f(\mathbf{x} + t\mathbf{u}) = \mathbf{a} \cdot (f'(\mathbf{x} + t\mathbf{u})(\mathbf{u})).$$

So by the mean-value theorem for 1-dimensional case, we get

$$F(1) - F(0) = F'(\theta). \quad (\exists \theta \in (0, 1))$$

Note that

$$F'(\theta) = \mathbf{a} \cdot (f'(\mathbf{x} + \theta\mathbf{u})(\mathbf{u})),$$

and

$$F(1) - F(0) = \mathbf{a} \cdot (f(\mathbf{y}) - f(\mathbf{x})).$$

Denote $\mathbf{z} \triangleq \mathbf{x} + \theta\mathbf{u}$, then we get the equation. ■

Corollary 1. If S is an open convex subset of $\mathbb{R}^n$, then equation (1) always holds.

Example 1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function, then we may take $a = 1 \in \mathbb{R}$, hence we get

$$f(\mathbf{y}) - f(\mathbf{x}) = f'(\mathbf{z})(\mathbf{y} - \mathbf{x}) = \nabla f(\mathbf{z}) \cdot (\mathbf{y} - \mathbf{x}).$$

Theorem 2. Let S be an open connected subset of $\mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}^m$ be a differentiable function on S . If $f'(\mathbf{x}) = \mathbf{0} \in L(\mathbb{R}^n; \mathbb{R}^m)$ is always a zero map on each point $\mathbf{x}$ of $\mathbb{R}^n$, then f is a constant function on S .

Proof: By the topological property of $\mathbb{R}^n$, since S is an open connected subset of $\mathbb{R}^n$, we know that for any two different points $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$, they can be joined by a polygonal arc in S , i.e., there is a map $\alpha : [0, 1] \rightarrow S$ such that $\alpha(0) = \mathbf{x}$ and $\alpha(1) = \mathbf{y}$ and the image $\alpha([0, 1])$ is a union of finite number of line segments. Denote the vertices of $\alpha([0, 1])$ to be $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_r$ with $\mathbf{p}_1 = \mathbf{x}$ and $\mathbf{p}_r = \mathbf{y}$, thus we know that $l(\mathbf{p}_i; \mathbf{p}_{i+1}) \subseteq S$ for all $i = 1, 2, \dots, r-1$. Using the mean-value theorem, we know that $\exists \mathbf{q}_i \in l(\mathbf{p}_i; \mathbf{p}_{i+1})$ such that

$$\mathbf{a} \cdot (f(\mathbf{p}_{i+1}) - f(\mathbf{p}_i)) = \mathbf{a} \cdot (f'(\mathbf{q}_i)(\mathbf{p}_{i+1} - \mathbf{p}_i)) = 0,$$

which holds for all $\mathbf{a}$. Hence, we have

$$\sum_{i=1}^{r-1} \left(\mathbf{a} \cdot (f(\mathbf{p}_{i+1}) - f(\mathbf{p}_i)) \right) = \mathbf{a} \cdot (f(\mathbf{y}) - f(\mathbf{x})) = 0.$$

Particularly, take $\mathbf{a} = f(\mathbf{y}) - f(\mathbf{x})$, see that $f(\mathbf{x}) = f(\mathbf{y})$ for any different arbitrary points $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, i.e., f is constant on S . ■

2 Sufficient Condition for Differentiability

• We have known that the existence of all partial derivatives or the existence of all directional derivatives cannot imply the differentiability of a vector-valued multivariable function. But the following theorem says that continuity of all but one of the partial derivatives can imply differentiability.

Theorem 3. (Sufficient Condition for Differentiability) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function, assume that one of the partial derivatives $D_1f, D_2f, \dots, D_nf$ exists at $\mathbf{x}_0$ and assume that the remaining $n-1$ partial derivatives exist in an n -ball $B(\mathbf{x}_0; r)$ and are continuous at $\mathbf{x}_0$. Then f is differentiable at $\mathbf{x}_0$.

Proof: Since $f \triangleq \begin{pmatrix} f_1 & f_2 & \dots & f_m \end{pmatrix}$ is differentiable $\iff$ each f_j is differentiable, so we only need to prove the case that f is real-valued function. So now suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Without loss of generality, we may assume that D_1f exists and the remaining continuous partial derivatives are $D_2f, D_3f, \dots, D_nf$. For any vector $\mathbf{v}$ in $\mathbb{R}^n$, write $\mathbf{v} \triangleq \lambda \mathbf{y}$ where $\|\mathbf{y}\| = 1$ and $\lambda = \|\mathbf{v}\|$, take λ small enough and hence $\mathbf{x}_0 + \mathbf{v} \in B(\mathbf{x}_0; r)$, in which $D_2f, D_3f, \dots, D_nf$ exist. Let $\{\mathbf{e}_j\}_{j=1}^n$ be the unit

coordinate vector for $\mathbb{R}^n$, then we can express $\mathbf{y}$ as

$$\mathbf{y} = \sum_{j=1}^n y_j \mathbf{e}_j, \quad (y_j \in \mathbb{R})$$

if we denote

$$\begin{aligned} \mathbf{v}_0 &:= \mathbf{0}, \\ \mathbf{v}_i &:= \sum_{j=1}^i y_j \mathbf{e}_j, \quad (i = 1, 2, \dots, n) \end{aligned}$$

then we get

$$f(\mathbf{x}_0 + \mathbf{v}) - f(\mathbf{x}_0) = f(\mathbf{x}_0 + \lambda \mathbf{y}) - f(\mathbf{x}_0) = \sum_{k=1}^n \left(f(\mathbf{x}_0 + \lambda \mathbf{v}_k) - f(\mathbf{x}_0 + \lambda \mathbf{v}_{k-1}) \right). \quad (2)$$

Note that the two points $\mathbf{x}_0$ and $\mathbf{x}_0 + \mathbf{v}_1 = \mathbf{x}_0 + \lambda y_1 \mathbf{e}_1$ differ only in the first component, since $D_1 f$ exists, we have

$$f(\mathbf{x}_0 + \lambda \mathbf{v}_1) - f(\mathbf{x}_0) = f(\mathbf{x}_0 + \lambda y_1 \mathbf{e}_1) - f(\mathbf{x}_0) = \lambda y_1 D_1 f(\mathbf{x}_0) + \lambda y_1 E_1(\lambda),$$

where $\lim_{\lambda \rightarrow 0} E_1(\lambda) = 0$; For $k \geq 2$, we denote

$$\mathbf{b}_k := \mathbf{x}_0 + \lambda \mathbf{v}_{k-1},$$

then

$$f(\mathbf{x}_0 + \lambda \mathbf{v}_{k-1} + \lambda y_k \mathbf{e}_k) - f(\mathbf{x}_0 + \lambda \mathbf{v}_{k-1}) = f(\mathbf{b}_k + \lambda y_k \mathbf{e}_k) - f(\mathbf{b}_k),$$

note that the two points $\mathbf{b}_k$ differ only in the k -th term, hence so we can also imply the one-dimensional mean-value theorem for derivatives, which results

$$f(\mathbf{b}_k + \lambda y_k \mathbf{e}_k) - f(\mathbf{b}_k) = \lambda y_k D_k f(\mathbf{a}_k). \quad (\exists \mathbf{a}_k \in l(\mathbf{b}_k; \mathbf{b}_k + \lambda y_k \mathbf{e}_k)) \quad (3)$$

Note that

$$\lim_{\lambda \rightarrow 0} \mathbf{b}_k = \lim_{\lambda \rightarrow 0} (\mathbf{x}_0 + \lambda \mathbf{v}_{k-1}) = \mathbf{x}_0,$$

hence

$$\lim_{\lambda \rightarrow 0} \mathbf{a}_k = \mathbf{x}_0.$$

By the condition, $D_k f$ is continuous at $\mathbf{x}_0$ for all $k \geq 2$, this implies that we can express $D_k f$ as

$$D_k f(\mathbf{a}_k) \triangleq D_k f(\mathbf{x}_0) + E_k(\lambda), \quad (4)$$

where $\lim_{\lambda \rightarrow 0} E_k(\lambda) = 0$. So insert the equation (4) into equation (3), we see that the equation (2) becomes

$$f(\mathbf{x}_0 + \mathbf{v}) - f(\mathbf{x}_0) = \lambda \sum_{k=1}^n y_k D_k f(\mathbf{x}_0) + \lambda \sum_{k=1}^n y_k E_k(\lambda) \triangleq \nabla f(\mathbf{x}_0) \cdot \mathbf{v} + \|\mathbf{v}\| E(\lambda), \quad (5)$$

where we denote

$$E(\lambda) := \sum_{k=1}^n y_k E_k(\lambda),$$

and hence

$$\lim_{\lambda \rightarrow 0} E(\lambda) = 0.$$

Thus the equation (5) means that f is differentiable at $\mathbf{x}_0$ by the definition of differentiability. ■

3 When Are The Mixed Partial Derivatives Equal?

- Consider the example below: define a real-valued function

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R},$$

$$(x, y) \mapsto \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then we see

$$D_1 f(x, y) = \frac{\partial}{\partial x} f(x, y) = \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, \quad \text{if } (x, y) \neq (0, 0),$$

and

$$D_1 f(0, 0) = 0,$$

thus we get

$$D_1 f(0, y) = -y \quad (\forall y \in \mathbb{R}),$$

and hence

$$D_{2,1} f(0, y) = -1,$$

so

$$D_{2,1} f(0, 0) = -1.$$

On the other hand, we compute

$$D_2 f(x, y) = \frac{\partial}{\partial y} f(x, y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, \quad \text{if } (x, y) \neq (0, 0),$$

and

$$D_2 f(0, 0) = 0,$$

so we know

$$D_2 f(x, 0) = 0 \quad (\forall x \in \mathbb{R}),$$

thus $D_{1,2} f(x, 0) = 1$ and hence

$$D_{1,2} f(0, 0) = 1,$$

which means that this two second-order partial derivatives at $(0, 0)$ do not equal, i.e.,

$$D_{1,2} f(0, 0) \neq D_{2,1} f(0, 0).$$

The following theorem gives a criterion for determining when the mixed partial derivatives equal.

Theorem 4. Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a function, if both partial derivatives $D_r f$ and $D_k f$ exist in an n -ball $B(\mathbf{x}_0; r)$ and if both $D_r f$ and $D_k f$ are differentiable at $\mathbf{x}_0$, then

$$D_{r,k} f(\mathbf{x}_0) = D_{k,r} f(\mathbf{x}_0).$$

Proof: If we write $f \triangleq (f_1 \ f_2 \ \dots \ f_m)$, then $D_k f = (D_k f_1 \ D_k f_2 \ \dots \ D_k f_m)$, which means that $D_{r,k} f(\mathbf{x}_0) = D_{k,r} f(\mathbf{x}_0) \iff D_{k,r} f_j = D_{r,k} f_j$ for all j . So it suffices to prove in the case that f is real-valued function, furthermore, we only need to consider the case when $n = 2$, which can guarantee the proof. Denote $\mathbf{x}_0 \triangleq (x_1, x_2) \in \mathbb{R}^2$, take an appropriate $h \in \mathbb{R}$ so that the four points $(x_1, x_2), (x_1 + h, x_2), (x_1, x_2 + h), (x_1 + h, x_2 + h)$ all lie in the 2-ball $B(\mathbf{x}_0; r)$. Denote

$$\Delta(h) := f(x_1 + h, x_2 + h) - f(x_1 + h, x_2) - f(x_1, x_2 + h) + f(x_1, x_2).$$

If we define $G(x) := f(x_1 + x, x_2 + h) - f(x_1 + x, x_2)$, then we see

$$\Delta(h) = G(h) - G(0).$$

Immediately, by 1-dimensional mean-value theorem, we get

$$\Delta(h) = hG'(x_0) = h \left(D_1 f(x_1 + x_0, x_2 + h) - D_1 f(x_1 + x_0, x_2) \right),$$

where $x_1 + x_0$ is between x_1 and $x_1 + h$. Note that we have

$$D_1 f(x_1 + x_0, x_2 + h) = D_1 f(x_1, x_2) + D_{1,1} f(x_1, x_2) x_0 + D_{2,1} f(x_1, x_2) h + (x_0^2 + h^2)^{\frac{1}{2}} E_1(h),$$

$$D_1 f(x_1 + x_0, x_2) = D_1 f(x_1, x_2) + D_{1,1} f(x_1, x_2) x_0 + |x_0| E_2(h),$$

where $\lim_{h \rightarrow 0} E_1(h) = \lim_{h \rightarrow 0} E_2(h) = 0$. Hence

$$\Delta(h) = D_{2,1} f(x_1, x_2) h^2 + E(h),$$

where we denote $E(h) := h(x_0^2 + h^2)^{\frac{1}{2}} E_1(h) - h|x_0| E_2(h)$. Note that $|x_0| < |h|$, so we get

$$0 \leq |E(h)| \leq \sqrt{2} h^2 |E_1(h)| + h^2 |E_2(h)|,$$

so we see that

$$\lim_{h \rightarrow 0} \frac{\Delta(h)}{h^2} = D_{2,1} f(x_1, x_2).$$

On the other hand, if we define $H(y) := f(x_1 + h, x_2 + y) - f(x_1, x_2 + y)$, then repeat the same process as above, we can find that

$$\lim_{h \rightarrow 0} \frac{\Delta(h)}{h^2} = D_{1,2} f(x_1, x_2),$$

which completes the proof. ■

Combining Theorem 3 and Theorem 4, we immediately get the following corollary.

Corollary 2. Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a function. If both partial derivatives $D_r f$ and $D_k f$ exist on an n -ball $B(\mathbf{x}_0; r)$ and if both $D_{r,k} f$ and $D_{k,r} f$ are continuous at $\mathbf{x}_0$, then

$$D_{r,k} f(\mathbf{x}_0) = D_{k,r} f(\mathbf{x}_0).$$